

helixllm-status

HelixLLM Full-Extension — §11.4.153 Comprehensive Per-Feature Status Ledger

Property	Value
Revision	1
Created	2026-07-11
Last modified	2026-07-11
Status	active — draft for Wave-3 commit to docs/features/Status.md
Track	(T1/feature/helixllm-full-extension - claude3)
Scope	§11.4.153 — every component/surface/feature of the HelixLLM full-extension programme, enumerated with Implementation / Wiring / Real-use / Tests-coverage / Validation / Evidence-path, reconciled against ACTUAL evidence on disk (§11.4.6 / §11.4.118).
Method	READ-ONLY filesystem audit. Base source: docs/qa/R41_EXTENSION_COVERAGE_LEDGER.md (2026-07-11 audit, 38 work-items) UPDATED against 13 newer gap-closing evidence directories committed later the same day (git log 8cd07e16..930cfddb). Every Evidence-path cell below was verified to exist on disk before being cited (ls/Read) — no row cites a fabricated or non-existent path.
Legend — Validation	PASS = real captured evidence exists and demonstrates the claim (§11.4.45/ §11.4.69/§11.4.107) · PENDING =

Property	Value
	implemented but no captured runtime evidence yet (§11.4.45 PENDING_FORENSICS-class) · SKIP = honestly not attempted this cycle (reason given) · OPERATOR-BLOCKED = genuinely blocked on an operator decision/resource (§11.4.21), not a quality gap.
Legend — Wiring	YES = end-user-reachable via a real network/CLI/API entry point today · PARTIAL = reachable via one path but not the full product path (e.g. raw script, not the broker) · NO = not wired into any user-reachable surface yet.

Anti-bluff notice (§11.4.6/§11.4.118/§11.4.153). A row is marked PASS ONLY when a real implementation citation (file:line/commit SHA) AND a real captured-evidence path that EXISTS ON DISK are both present. Where evidence is stale, partial, or the artifact was regenerated by a substitute (e.g. a non-production model), that gap is stated explicitly in the Real-use or a footnote — never silently upgraded to PASS.

1. Serving & VRAM Broker

#	Component	Feature	Category	Implementation	Wi
1	HelixLLM serving	Lane-A coder serving (Qwen3-Coder-30B-A3B, :18434)	Serving	live container helixllm-coder, broker ClassCoder (submodules/helix_llm/internal/vrambroker/broker.go:19)	YE HT ch co PI res
2	HelixLLM serving	Lane-B 2nd instance	Serving	broker.go:30 ClassAgent; cmd/agentgen-boot/main.go (c92fb16)	YE res

#	Component	Feature	Category	Implementation	Wi
		(Mistral-Nemo-12B, cmd/agentgen-boot, ClassAgent)			ber tea cyc
3	HelixLLM serving	VRAM broker CORE (7 classes, fail-closed admission)	Serving	broker.go:18-36, admit() :189	YE GP cap bo led thr
4	HelixLLM serving	Pre-boot lane-config validator (cmd/laneconfig-validate)	Serving	validate.go KV formula :154, -ngl :262, port/name collision :269-281	YE eve bo
5	HelixLLM serving	D6 tool-calling-under-load guard (array-as-string #1809 bug class)	Serving/ Reliability	internal/gateway/ tool_call_normalizer.go, internal/brain/ openai_provider.go:213-262; new permanent guard helix_code/ internal/llm/ tool_calling_concurrent_test.go	YE cod N= cor
6	HelixLLM serving	GGUF-revision verification (running quant postdates Unsloth tool-calling fix)	Serving/ Provenance	N/A (research + provenance check, no code)	N/A

#	Component	Feature	Category	Implementation	Wi
7	HelixLLM serving	Phase-4 Lane-A kv-unified/ batch-size/ parallel tuning (Tasks 1.2/1.3/4.1)	Serving/ Perf	pause mechanism proven; flag changes NOT applied	NO

2. Capabilities

#	Component	Feature	Category	Implementation
8	HelixLLM capability	Vision-VLM (Qwen2.5-VL-3B, :18439)	Capability	cmd/visiongen-boot/main.go (Phase
9	HelixLLM capability	Image-gen (FLUX.1-schnell, :18442, ClassImage)	Capability	scaffold 0f07559; live path 722cf05
10a	HelixLLM capability	Video-gen liveness re-analysis (WAN2.2 TI2V-5B clip, vidanalyzer)	Capability	submodules/helix_llm/docs/qa/ phase4_videogen_20260707/harnes vidanalyzer/main.go — advancedDif recalibrated 1.5 → 0.55 on real fixture (\$

#	Component	Feature	Category	Implementation
10b	HelixLLM capability	Video-gen broker-integrated runtime path (cmd/videogen-boot → ClassVideo admission)	Capability	scaffold 9145505; README self-declared captured runtime proof”
11	HelixLLM capability	Vectorization (raster → SVG, vtracer default path)	Capability	NEW submodules/helix_llm/service vectorize/ (Go shim + real vtracer 0 rsvg-convert)
11b	HelixLLM capability	Vectorization — StarVector-8B optional GPU tier	Capability	not attempted
12	HelixLLM capability	Translation (NLLB-200, :18436)	Capability	e58aab9f
13	HelixLLM capability	Whisper STT (:18437)	Capability	e1b3b41
14			Capability	f8632e3

#	Component	Feature	Category	Implementation
	HelixLLM capability	Tesseract OCR (:18438)		
15	HelixLLM capability	Embeddings (bge-small via TEI)	Capability	55bdf9b6
16	HelixLLM capability	RAG-TEI core (in-memory embed → cosine-retrieve → ground, :18440)	Capability	harness submodules/helix_llm/docs/phase3_rag_20260707/harness/ (87
17	HelixLLM capability	RAG Qdrant vector-DB + cross-encoder reranker fusion	Capability	NEW submodules/helix_llm/docs/phase3_rag_qdrant_rerank_20260707/harness/ — real Qdrant ANN + TEI / reranker-base)
18	HelixLLM capability	HelixMemory (durable memory)	Capability	helix_code/internal/memory/helixmemory_provider.go (wraps digital.vasic.helixmemory/pkg/1
19	HelixLLM capability	cognee wiring	Capability	pre-existing, disabled cognee_integra
20	HelixLLM capability	Network-provider (LAN/VPN endpoint)	Capability	landed per programme RESUME; bank helixllm_network_provider.yaml

3. Protocols & Core Providers

#	Component	Feature	Category	Implementation
21	HelixCode server	Dual-wire facade (OpenAI /v1/chat/completions + Anthropic /v1/messages)	Protocol	helix_code/internal/server/server.go route grp + wireFacadeAuthMiddleware :594 (8a309004)
22	HelixLLM protocol	MCP gateway (real go-sdk v1.6.1, Streamable-HTTP)	Protocol	submodules/helix_llm/cmd/gateway + internal/mcpgat (895788c5)
23	HelixLLM protocol	A2A real wire (Google A2A, v0.3.15 SDK)	Protocol	submodules/helix_llm/int a2a/* + cmd/a2a-server/ma
24	HelixAgent	Legacy proprietary ACP stub retirement	Protocol	stub STILL PRESENT unchang submodules/helix_agent/c api/main.go /api/v1/acp/ {execute,broadcast,statu (canned responses)
25a	Dev tooling	codegraph wiring fix (path-rot + exclude-bloat)	Tooling	path-rot 7e8e4946; exclude-b bc89f2ac (51,961 → 22,548 file
25b	Dev tooling	codegraph — committed post-fix codegraph_validate.sh re-run	Tooling	same fix as 25a
26	Dev tooling	OpenDesign bring-up (: 7456, helixcode-brand seed)	Tooling	daemon built off-tree .opend src; .mcp.json enable c1fe
27	LLMsVerifier	C1 → C5 resolver chain	Provider infra	submodules/llms_verifier registry_resolve.go (167L
28	Provider config	Extended-provider config rows (13: poe/perplexity/sakana/hunyuan/xai/	Provider infra	providers/config.go (c696

#	Component	Feature	Category	Implementation
		moonshot/zai/fireworks/ deepinfra/ai21/reka/ hyperbolic/xiaomi)		
29	HelixCode LLM	CONST-039 per-provider live-proof harness	Provider infra	helix_code/internal/llm/provider_live_proof_test (424L, tag providerlive)
30	Provider catalogue	Catalogue-providers (HF-router + together)	Provider infra	helix_code/internal/llm/openai_compatible_catalog (7210f373)
31	LLMsVerifier	HelixLLM-as-tracked-provider registration	Provider infra	submodules/llms_verifier 760bfde9
32	claude_toolkit	Provider-alias reconciliation	Dev tooling	claude_toolkit 5d611d9 (+ fireworks-ai/deepinfra/moons
33	claude_toolkit	R5 port-ownership hardening	Dev tooling	claude_toolkit ef77b19 (detect_helixagent_recom merged to main
34	Provider config	Dead-code disposition (cerebras/hf/together/replicate)	Provider infra	investigation only, 0 prod refs confirmed

4. HelixQA Banks / Challenges / Race Coverage (extension test infrastructure)

#	Component	Feature	Category	Implementation
35	HelixQA	Coder bench + concurrency banks	Test infra	helixllm_coder_bench.yaml(7)/helixllm_coder_concurrency.yaml(6)
36	HelixQA	Coder DDoS + chaos + memory banks	Test infra	helixllm_coder_ddos.yaml/_chaos.yaml; cmd/helixqa-verify-coder-chaos patched to source model from HELIX_CODER_MODEL

#	Component	Feature	Category	Implementation
37	HelixQA	Coder race/ deadlock bank (LLM coder, :18434)	Test infra	NEW banks/helixllm_coder_race.yaml (7 cases) + NEW cmd/helixqa-verify- coder-race-llm/main.go
37b	HelixQA	HelixCode HTTP-server ModelManager race bank (:8080, distinct scaffold)	Test infra	banks/helixcode_coder_race.yaml + cmd/helixqa-verify-coder-race (cb3ce95f)
38	Challenges submodule	HelixLLM coder live end- to-end anti- bluff Challenge	Test infra	NEW submodules/challenges/ challenges/scripts/ helixllm_coder_live_e2e_challenge.
39	HelixCode inner module	\$11.4.169 -race detector coverage sweep (9 concurrency- sensitive package trees)	Test infra	new permanent guard helix_code/ internal/llm/ cross_provider_registry_race_test.
40	HelixCode inner module	Pre-existing full-suite test- failure fix (internal/ security, internal/ tools/ browser)	Test infra	internal/security/translator_test. reconciled (\$11.4.120); internal/tools/ browser/{controller.go,actions.go} real deadline-precedence bug fixed
41	HelixCode inner module	Full codebase test-suite run	Test infra	N/A (dynamic confirmation pass)

#	Component	Feature	Category	Implementation
		(go build/go vet/go test)		

5. Governance / Security

#	Component	Feature	Category	Implementation	Wiring	Real-use
42	Provider evidence	Committed Gemini API-key leak (CONST-042)	Security	N/A (incident)	N/A	Real secret-shape in docs/qa/phase1_provider_live_probe.md; discovered by an pass, NOT a pre-e escape)

6. Honest coverage summary (§11.4.6 / §11.4.118)

- **43 rows enumerated** (38 base R41 items, several split into sub-rows for honesty — e.g. video-gen 10a/10b, codegraph 25a/25b, race-bank 37/37b — plus 4 new rows for work done in this session's later commits: D6/GGUF closure evidence citation, the standalone -race sweep, the pre-existing test-failure fix, the full-suite dynamic run, and the Gemini key-leak incident).
- **34 PASS** — real captured evidence exists on disk and was verified to exist before citing.
- **6 OPERATOR-BLOCKED** — genuinely blocked on an operator decision/resource (coder-pause window, API keys, cognee upstream bug, ACP stub disposition, dead-code disposition, secret rotation), not a quality gap.
- **3 PENDING / honest gaps** — video-gen broker-integrated path (10b), codegraph post-fix codegraph_validate.sh committed re-run (25b), HelixCode-server :8080 ModelManager race bank (37b).
- **1 SKIP (honest defer)** — StarVector-8B optional GPU tier (11b) — VRAM headroom insufficient, documented follow-up.

This ledger **supersedes** docs/qa/R41_EXTENSION_COVERAGE_LEDGER.md's headline figures for 6 items that were gap-closed by later commits the same day (D6/GGUF #5-#6, RAG-Qdrant #17, Vectorization #11, HelixQA DDoS/chaos/memory #36, HelixQA race bank #37, Challenges #38, OpenDesign durable evidence #26) —

those items moved from GAP/OPERATOR-GATED states to DONE-PROVEN/PASS with real evidence citations that were verified to exist on disk during this audit. The R41 document remains the authoritative historical audit trail; this document is the current, reconciled state.

No row in this ledger cites a fabricated evidence path. Every docs/qa/<run-id>/ path above was confirmed present via ls/Read during this audit session.

Sources verified

- docs/qa/R41_EXTENSION_COVERAGE_LEDGER.md (base audit, read in full)
- 17 evidence directories read in full or verified present:
 - rag_qdrant_liveproof_20260711T142237Z,
 - vectorization_liveproof_20260711T142839Z,
 - race_sweep_20260711_142529,
 - helixllm_challenge_20260711T141755Z,
 - videogen_recalibration_20260711_144236Z,
 - videogen_analysis_20260711T141601Z,
 - opendesign_durable_evidence_20260711T141015Z,
 - a2a_wireshape_fix_20260711T140403Z,
 - a2a_live_e2e_20260711T134958Z,
 - helixqa_gapfill_live30b_20260711T140607Z,
 - d6_toolcalling_gapfill_20260711T140055Z,
 - cpucaps_liveproof_20260711T134045Z,
 - dualwire_live_e2e_20260711T130459Z,
 - generative_liveproof_20260711T131205Z,
 - lane_b_liveproof_20260711T130533Z,
 - mcp_gateway_live_e2e_20260711T132250Z,
 - rag_liveproof_20260711T133918Z,
 - vision_liveproof_20260711T133823Z,
 - helixqa_live_vnv_20260711T130200Z,
 - coder_boot_liveproof_20260711T125759Z,
 - R41_TEST_SUITE_RUN_20260711T135818Z.md,
 - R41_testfail_fix_20260711T141510Z.md,
 - SECURITY_INCIDENT_gemini_key_leak_20260711.md.
- git log --oneline -40 (root) and git -C submodules/helix_llm log --oneline -25 — cross-referenced commit SHAs cited above against the actual git history.